# Temmam Nabil Altaie

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## Education

**Norwegian University of Science and Technology (NTNU)** — Trondheim, Norway
*Bachelor of Engineering, Electronic Systems (after preparatory science course, 2023–2024)* — *Aug 2024 – Jun 2027*

- Grade A in all four engineering project courses
- Completed: Electronics Design, Electronic Systems Design and Implementation I–III, Analog and Digital Electronics, Computer Networks for Instrumentation
- Current: Embedded Systems, Advanced Sensor Systems, Radio Communication (RF analysis with the Nordic nRF54L15 as case), bachelor's thesis

**Vest-Lofoten Upper Secondary School** — Leknes, Norway
*Sports and Physical Education* — *Aug 2018 – Jun 2021*

## Projects

**Wearable Object Detection for the Blind** | *Python, YOLOv8, Raspberry Pi, ESP32* — Jan 2026 – Jun 2026

- Led software in a team of three; real-time YOLOv8n object detection on a Raspberry Pi 5 at 6–8 FPS
- Mapped object position to seven zones, sent over UDP to an ESP32-C3 driving a haptic belt in under 100 ms
- Added offline Norwegian speech output (Piper) and speech recognition (Whisper) in multithreaded Python

**Bare-Metal Peripheral Drivers for STM32F411** | *C, ARM Cortex-M4, STM32CubeIDE* — Sep 2025 – Oct 2025

- Wrote register-level drivers for GPIO, $I^2C$, SPI, USART and clock configuration without vendor HAL libraries, about 4 700 lines of C including a custom device header with register maps
- Implemented both blocking and interrupt-driven transfers with NVIC configuration and callback events; verified with test applications

**ECG Measurement System** | *LTspice, KiCad, ESP32, C++, JavaScript* — Aug 2025 – Nov 2025

- Designed an analog front end with 1000 V/V gain and 0.03–16 Hz bandwidth; verified in LTspice and on prototype
- Implemented R-peak detection on an ESP32 with a 16-bit $I^2C$ ADC; heart rate within 1–3 BPM of a reference
- Streamed live data over WebSocket to a web dashboard; four-layer PCB in KiCad

**Offline Voice Assistant for Smart Homes** | *ESP32, Python, Docker, Home Assistant* — Jan 2025 – Jun 2025

- Streamed $I^2S$ microphone audio from an ESP32 over WebSocket to local speech-to-text, triggering Home Assistant automations without cloud services; under 5 s response and up to 90 % accuracy

## Technical Skills

**Programming**: C, C++, Python, JavaScript, HTML, LaTeX
**Hardware**: ESP32, STM32, AVR, Raspberry Pi, analog filter and amplifier design, PCB layout
**Interfaces**: $I^2C$, SPI, $I^2S$, UART, PWM, UDP, WebSocket, Wi-Fi
**Tools**: KiCad, LTspice, Git, Linux, Docker, oscilloscope, signal generator
**Languages**: Norwegian (native), English (fluent), Arabic

## Experience

**Learning Assistant** — Jan 2026 – Present
*NTNU, Department of Electronic Systems* — *Trondheim, Norway*

- Supervise lab and project work in Electronic System Design and Implementation I and II
- Planned and taught crash courses in C programming and analog filter design for about 17 students

**Substitute Teacher** — Jan 2023 – Jun 2023
*Leknes Lower Secondary School* — *Leknes, Norway*

**Conscript Service, Signal Horn Player** — Sep 2021 – Sep 2022
*His Majesty the King's Guard, 3rd Company (Band and Drill)* — *Oslo, Norway*

- Signal horn in the music troop of a unit built on discipline, order and precision; assignments included the Platinum Jubilee in Windsor, a Balkan tour, military tattoos in Basel and Oslo, and events for heads of state

**Part-Time Employee** — Jun 2021 – Present
*Deli de Luca (Esso station), seasonal* — *Leknes, Norway*

**Founder and Trumpet Player** — Jan 2018 – Present
*Trompet ved Altaie (sole proprietorship)* — *Norway*

- Run a small business performing at weddings, funerals and church services; handle clients and marketing

## Leadership

**Vice President** — Jan 2026 – Present
*NTNUI Calisthenics* — *Trondheim, Norway*

- Share responsibility for a club of 200+ athletes and coordinate a team of 14 coaches
- Planned a training camp in Spain for 40 participants and wrote funding applications to the NTNUI board

**Class Representative and Reference Group Member** — Aug 2024 – Present
*NTNU, Electronic Systems* — *Trondheim, Norway*

- Represent the class in regular meetings with faculty and teaching staff to improve courses